

Higher projective maximal orthogonal families of measures

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Abstract

Let $1 \leq n < \omega$ and assume that the canonical mouse M_n with n Woodin cardinals exists. We describe a forcing extension of M_n in which every boldface Σ_{n+2}^1 set of reals has the Baire property, boldface Σ_{n+2}^1 -uniformization holds, and there is a lightface Π_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω . Additionally there is no boldface Σ_{n+2}^1 maximal orthogonal family in the same model.

1 Introduction

Let $P(2^\omega)$ be the space of Borel probability measures on Cantor space. Two measures μ and ν are orthogonal, written $\mu \perp \nu$, if they are concentrated on disjoint Borel sets. A maximal orthogonal family, abbreviated *mof*, is a maximal set of pairwise orthogonal measures.

The study of definable mofs begins with a gap between the two sides of the first projective level. No analytic mof exists, while Fischer and Törnquist proved that $V = L$ implies the existence of a lightface Π_1^1 mof [FT10]. Fischer, Friedman, and Törnquist later obtained the corresponding separation at the second projective level. They constructed extensions of L with a lightface Π_2^1 mof and no boldface Σ_2^1 mof. Their models also contain a Δ_3^1 well-order of the reals, and the construction works both when the continuum is $\aleph_2$ and when it is larger [FFT12, Theorems 1 and 2].

The latter result leaves open the following natural higher-level placement problem. For a fixed projective level $m \geq 1$, is it consistent that

$$\text{there is a lightface } \Pi_m^1 \text{ mof,} \quad \text{but} \quad \text{there is no boldface } \Sigma_m^1 \text{ mof?} \quad (\mathbf{P}_m)$$

This asks for more than the existence of a projectively definable mof. It asks that a mof be placed on the Π -side of a prescribed projective level while no mof belongs to the corresponding Σ -pointclass. Problems of this form are also studied for projective well-orders of the reals. There one seeks models in which a well-order first has a definition at a prescribed projective level, while definitions of lower complexity are ruled out. The constructions of projective well-orders and definable maximal families have therefore developed in parallel and use related coding methods [FFT12, FZ11].

The present paper answers $(\mathbf{P}_m)$ at every higher finite projective level, relative to the corresponding canonical mouse. More precisely, if $1 \leq n < \omega$ and M_n exists, then there is a forcing

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extension of M_n with a lightface Π_{n+2}^1 mof and no boldface Σ_{n+2}^1 mof. Thus, for $m \geq 3$, the existence of M_{m-2} gives the consistency of (P_m) . Together with the results at levels 1 and 2 described above, this gives the corresponding relative consistency statement at every finite projective level. The restriction $n \geq 1$ in the theorem below reflects the use of the least Woodin cardinal of M_n , the weakly homogeneous tree T_{n+1} , and Steel's analysis of $M_n(s)$. The earlier arguments at levels 1 and 2 use L and a different ground-model analysis.

We prove the following theorem.

Theorem 1.1. *Let $1 \leq n < \omega$ and assume that M_n exists. There is a cardinal-preserving forcing extension V^* of M_n such that:*

- (1) $V^* \models \text{CH}$;
- (2) every boldface Σ_{n+2}^1 set of reals has the Baire property;
- (3) boldface Σ_{n+2}^1 -uniformization holds;
- (4) there is a lightface Π_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω ;
- (5) there is no boldface Σ_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω .

The proof separates the positive and negative parts of the placement problem. For the positive part, we use a finite-support iteration with mof stages and category amoeba stages. At a mof stage, a real f coding a measure is considered. If the measure is orthogonal to all measures accepted earlier, the real $\langle \text{mof}, f \rangle$ is written into a fresh block of independent Suslin trees. A self-indexed almost disjoint real u selects the active columns of the block and codes localizations of the resulting branch pattern. The Fischer–Törnquist coding map then replaces f by a code for an equivalent measure from which $f \oplus u$ can be recovered [FT10, FFT12]. The use of fresh blocks gives exact decoding, and the local form of the code yields a lightface Π_{n+2}^1 definition of the final family.

For the negative part, category amoeba stages add Borel meager sets which cover the Borel meager sets coded in prescribed inner models. Together with projective correctness for small generic extensions of M_n , this gives the Baire property for every boldface Σ_{n+2}^1 set. The uniformization statement does not come from the iteration itself. If N is a small generic extension of M_n and $s \in \mathbb{R}^N$, then the canonical mouse $M_n(s)$ carries a good $\Sigma_{n+2}^1(s)$ well-order of its reals. Steel capture places witnesses for $\Sigma_{n+2}^1(s)$ statements into $M_n(s)$, and choosing the least witness in this order gives a uniformizing relation of the same complexity. Finally, the Baire property and uniformization at the Σ_{n+2}^1 level rule out a Σ_{n+2}^1 mof by the Kechris–Sofronidis map and Hjorth turbulence.

All pointclasses in the regularity and uniformization statements are boldface. The mof constructed in the proof is lightface definable. We use standard recursive pairings of reals without further comment. Fix once and for all a recursive real $\tau_{\text{mof}} \in 2^\omega$, for example 0^ω , and write mof for this fixed tag at mof stages.

2 Measures and orthogonality

We recall the facts about measures used below. See [FT10, FFT12] for details.

Let

$$p(2^\omega) = \{f : 2^{<\omega} \longrightarrow [0, 1] : f(\emptyset) = 1 \text{ and } f(s) = f(s \smallfrown 0) + f(s \smallfrown 1)\}.$$

For $f \in p(2^\omega)$, let μ_f be the measure determined by

$$\mu_f(N_s) = f(s),$$

where $N_s = \{x \in 2^\omega : s \subseteq x\}$. This gives a recursive presentation of $P(2^\omega)$. Let

$$p_c(2^\omega) = \{f \in p(2^\omega) : \mu_f \text{ is non-atomic}\}.$$

The set $p_c(2^\omega)$ is arithmetical. The relations

$$\mu_f \perp \mu_g, \quad \mu_f \ll \mu_g, \quad \mu_f \approx \mu_g$$

are arithmetical on measure codes.

Lemma 2.1 (Fischer–Törnquist coding lemma). *There are an arithmetical partial decoding map*

$$r_{\text{FT}} : \text{dom}(r_{\text{FT}}) \longrightarrow 2^\omega$$

whose domain and graph are Π_1^0 , and a recursive function

$$G_{\text{FT}} : p_c(2^\omega) \times 2^\omega \longrightarrow p_c(2^\omega)$$

such that, for every $f \in p_c(2^\omega)$ and $u \in 2^\omega$,

$$\mu_{G_{\text{FT}}(f,u)} \approx \mu_f \quad \text{and} \quad r_{\text{FT}}(G_{\text{FT}}(f,u)) = u.$$

This is [FT10, Coding Lemma 3.5 and Remark 3.6]. The function G_{FT} changes conditional weights along a recursive sequence of splitting nodes. The bits of u are read from the resulting ratios. Absolute equivalence with the original measure is preserved.

We also use the Kechris–Sofronidis map. Define

$$xE_Iy \iff \sum_{k < \omega} \frac{|x(k) - y(k)|}{k+1} < \infty.$$

Fact 2.2 (Kechris–Sofronidis). *There is a Borel map $x \mapsto \mu^x$ from 2^ω to $P(2^\omega)$ such that*

$$xE_Iy \implies \mu^x \approx \mu^y, \quad \neg(xE_Iy) \implies \mu^x \perp \mu^y.$$

The equivalence relation E_I is turbulent and each E_I -class is meager [KS01, Hjo00].

For a fixed measure μ , the relation $\ll$ has the ccc below μ : every family of pairwise orthogonal nonzero measures which are absolutely continuous with respect to μ is countable. We use this in the proof that no Σ_{n+2}^1 mof exists.

3 The input from M_n

Fix $1 \leq n < \omega$, and let δ_0 be the least Woodin cardinal of M_n . We isolate the part of the analysis of M_n used below. The lower-part analysis is recorded in the two companion papers [Hof26, HM26, for the Π -side, see Fact 2.12, Lemmas 2.13 and 2.15, Theorem 2.18, Fact 2.19, and Lemma 2.20]. The weak-homogeneity and small-generic absoluteness statements are [Hof26, HM26, for the Π -side, see Theorem 2.23, Corollary 2.24, Theorem 2.25, and Lemma 2.27]; see also [Ste95, Ste10, MS08].

Definition 3.1. In an ω_1 -preserving extension of M_n arising below, let $\mathcal{I}_n$ be the class of countable premice $\mathcal{M}$ such that:

- (1) $\mathcal{M}$ is a passive ordinary lower-part premouse, that is, it has no Woodin cardinal;
- (2) $\mathcal{M}$ is n -small and ω -sound, and

$$\rho_\omega(\mathcal{M}) = \omega;$$

- (3) $\mathcal{M}$ satisfies Steel's Π_n -iterability condition.

Premice are identified with their standard real codes.

Theorem 3.2. The following statements hold in M_n and in every ω_1 -preserving forcing extension of M_n arising in the construction below.

- (1) The class $\mathcal{I}_n$ is Π_{n+1}^1 in the codes. Every member of $\mathcal{I}_n$ is of the form $\mathcal{J}_\eta^{M_n}$ for some $\eta < \omega_1$, and such η are cofinal in ω_1 . Consequently,

$$m_\infty = \bigcup \mathcal{I}_n = \mathcal{J}_{\omega_1}^{M_n}.$$

- (2) There are canonical sequences

$$\vec{S} = \langle S_\xi : \xi < \omega_1 \rangle \quad \text{and} \quad \vec{d} = \langle d_{\xi, m} : \xi < \omega_1, m < \omega \rangle$$

which are uniformly definable over m_∞ without real parameters. The sequence $\vec{S}$ is independent: for every finite $e \subseteq \omega_1$,

$$\prod_{\xi \in e} S_\xi$$

is a Suslin tree. The sets $d_{\xi, m}$ are infinite and pairwise almost disjoint.

If $\mathcal{M} \in \mathcal{I}_n$, then $\mathcal{M}$ computes the restrictions of $\vec{S}$ and $\vec{d}$ below $\omega_1^{\mathcal{M}}$, and these computations agree for different members of $\mathcal{I}_n$.

- (3) There is a canonical tree $T_{n+1} \in M_n$ such that

$$p[T_{n+1}]$$

is the universal Σ_{n+2}^1 set. For every $\kappa < \delta_0$, the tree T_{n+1} is κ -weakly homogeneous in M_n . If $\mathbb{Q} \in M_n$ has M_n -cardinality less than δ_0 , $H \subseteq \mathbb{Q}$ is M_n -generic, and $a \in \mathbb{R}^{M_n[H]}$, then

$$L[T_{n+1}, a] \prec_{\Sigma_{n+2}^1} M_n[H],$$

where elementarity concerns projective formulas with real parameters from $L[T_{n+1}, a]$.

- (4) $M_n \models \text{CH}$.

Proof. All clauses in Definition 3.1 except iterability are arithmetical in a premouse code. Steel's analysis shows that Π_n -iterability is Π_{n+1}^1 . Comparison of a member of $\mathcal{I}_n$ with a realizable lower-part initial segment of M_n shows that it is itself an initial segment of M_n . Conversely, cofinally many initial segments of $\mathcal{J}_{\omega_1}^{M_n}$ satisfy Definition 3.1. This proves (1).

The canonical construction order of M_n , together with condensation, gives a canonical $\diamond_{\omega_1}$ -sequence over m_∞ . The usual recursion from this sequence gives the independent Suslin sequence

$\vec{S}$. The same construction order gives the almost disjoint matrix $\vec{d}$. The definitions commute with passage to members of $\mathcal{I}_n$, which proves (2).

The Martin–Steel scale analysis gives in M_n a weakly homogeneous tree representing the universal Σ_{n+2}^1 set. We choose the least such tree in the construction order. Given a forcing of size below δ_0 , choose a completeness bound $\kappa < \delta_0$ above its size. The Martin–Solovay complementing-tree theorem then gives the displayed small-generic correctness. This proves (3). Clause (4) follows from the fine structure of M_n . $\square$

4 Coding with fresh Suslin-tree blocks

This section defines the coding operation used at mof stages. The use of $\Delta(u)$ follows the coding method in [FFT12, FFZ11].

4.1 Columns in one fresh block

For $\eta < \omega_1$, let

$$\vec{S}^\eta = \langle S_{\omega\eta+j} : j < \omega \rangle.$$

Fix a recursive bijection

$$\rho : (\omega \setminus \{0\}) \times \omega \times 2 \longrightarrow \omega.$$

For $m > 0$, $k < \omega$, and $i < 2$, put

$$S_{k,i}^{\eta,m} = S_{\omega\eta+\rho(m,k,i)}.$$

Thus the block $\vec{S}^\eta$ is split into columns indexed by the positive integers. Each column is split into pairs

$$\{S_{k,0}^{\eta,m}, S_{k,1}^{\eta,m}\} \quad (k < \omega).$$

Let $z \in 2^\omega$. For $m > 0$, define

$$\mathbb{B}_{\eta,m}(z) = \prod_{k < \omega}^{\text{fin}} S_{k,z(k)}^{\eta,m}.$$

If $H_m \subseteq \mathbb{B}_{\eta,m}(z)$ is generic, let

$$b_{m,k} = \bigcup \{p(k) : p \in H_m \text{ and } k \in \text{dom}(p)\}.$$

Then $b_{m,k}$ is a cofinal branch through $S_{k,z(k)}^{\eta,m}$.

Let

$$\mathbb{B}_\eta^*(z) = \prod_{m > 0}^{\text{fin}} \mathbb{B}_{\eta,m}(z).$$

Equivalently, this is the finite-support product of the trees

$$S_{k,z(k)}^{\eta,m} \quad (m > 0, k < \omega).$$

Lemma 4.1. *The forcing $\mathbb{B}_\eta^*(z)$ is ccc. It preserves every finite product of members of $\vec{S}$ none of which is one of the selected trees $S_{k,z(k)}^{\eta,m}$.*

Proof. There are countably many finite supports. On a fixed support, the forcing is a finite product of distinct members of $\vec{S}$. Such a product is ccc by independence. For preservation, apply the same argument after adjoining the given finite product of unselected trees. $\square$

4.2 Localization of one column

Fix $m > 0$, and let

$$\mathcal{B}_{\eta,m,z} = \langle b_{m,k} : k < \omega \rangle$$

be the branch sequence added by $\mathbb{B}_{\eta,m}(z)$. We first package this sequence into a transitive model of size $\aleph_1$ and then code that model by a subset of ω_1 .

Recall that

$$m_\infty = \mathcal{J}_{\omega_1}^{M_n} = \bigcup \mathcal{I}_n.$$

In the branch extension, let $\lambda_{\eta,m,z}$ be least such that

$$N_{\eta,m,z} = L_{\lambda_{\eta,m,z}}[m_\infty, \eta, m, \mathcal{B}_{\eta,m,z}]$$

has size $\aleph_1$ and satisfies:

- (1) $N_{\eta,m,z} \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$ and $\omega_1^{N_{\eta,m,z}} = \omega_1$;
- (2) $N_{\eta,m,z}$ contains m_∞ , η , m , and $\mathcal{B}_{\eta,m,z}$;
- (3) using m_∞ , it computes $\vec{S}$ and $\vec{d}$ and verifies that

$$b_{m,k} \text{ is a cofinal branch through } S_{k,z(k)}^{\eta,m} \quad (k < \omega).$$

Thus z is read from the branch pattern and is not included as a separate distinguished object.

Let

$$X_{\eta,m,z} \subseteq \omega_1$$

be the canonical code for

$$(N_{\eta,m,z}, \in, m_\infty, \eta, m, \mathcal{B}_{\eta,m,z}).$$

More precisely, we use the fixed canonical enumeration of $N_{\eta,m,z}$ of length ω_1 , with the displayed objects as distinguished constants, and code the resulting well-founded extensional relation. Hence the transitive collapse and the distinguished objects are recovered from $X_{\eta,m,z}$. The component coding the block address η is fixed independently of the column. Its initial segments are used in every localization. Consequently, different columns belonging to the same block recover the same local block address.

We reshape this code as follows. Choose the least λ of size $\aleph_1$ such that

$$L_\lambda[X_{\eta,m,z}] \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}.$$

Inside this model, take the least continuous increasing sequence

$$\langle M_\xi : \xi < \omega_1 \rangle$$

of countable elementary submodels whose union is $L_\lambda[X_{\eta,m,z}]$, and put

$$c_\xi = M_\xi \cap \omega_1, \quad C = \{c_\xi : \xi < \omega_1\}.$$

The set $Y_{\eta,m,z} \subseteq \omega_1$ codes $X_{\eta,m,z}$ on the odd ordinals. Its even part codes the increasing enumeration of C by the fixed interval pattern: before c_0 it codes a well-order of type c_0 , and between c_ξ and $c_{\xi+1}$ it codes a well-order of type $c_{\xi+1}$.

We now isolate the first-order property expressed by this code. Let M be a transitive model of

$$\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”},$$

let $\mathcal{M} \in \mathcal{I}_n$ satisfy

$$\omega_1^{\mathcal{M}} = \omega_1^M,$$

and let $Y \subseteq \omega_1^M$. We write

$$M \models \text{LocalCode}_n(Y, z, \mathcal{M}, \bar{\eta}, m)$$

if the following first-order assertion holds in M . The odd part of Y , together with the reshaping information on its even part, codes a well-founded extensional structure whose transitive collapse $\bar{N}$ satisfies:

- (1) $\bar{N} \models \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”}$ and has size $\aleph_1^M$;
- (2) $\bar{N}$ contains the distinguished objects $\mathcal{M}, \bar{\eta}, m$, and a sequence

$$\langle \bar{b}_k : k < \omega \rangle;$$

- (3) using $\mathcal{M}$, the model $\bar{N}$ computes the local versions of $\vec{S}$ and $\vec{d}$;
- (4) for every $k < \omega$, $\bar{b}_k$ is a cofinal branch of length ω_1^M through

$$S_{\omega\bar{\eta}+\rho(m,k,z(k))}^{\mathcal{M}}.$$

Here $S_{\xi}^{\mathcal{M}}$ denotes the ξ -th tree computed from $\mathcal{M}$. Thus the localized model reads the same real z from one column of one block.

Lemma 4.2 (Localization). *Let M be a countable transitive model of*

$$\text{ZFC}^- + \text{“}\aleph_1 \text{ exists”},$$

and let $z, \mathcal{M} \in M$, where

$$\mathcal{M} \in \mathcal{I}_n \quad \text{and} \quad \omega_1^{\mathcal{M}} = \omega_1^M.$$

If

$$Y_{\eta,m,z} \cap \omega_1^M \in M,$$

then, for some $\bar{\eta} < \omega_1^M$,

$$M \models \text{LocalCode}_n(Y_{\eta,m,z} \cap \omega_1^M, z, \mathcal{M}, \bar{\eta}, m).$$

If the data arise by collapsing an elementary hull containing η , then $\bar{\eta}$ is the image of η under the collapse.

Proof. The even part of $Y_{\eta,m,z} \cap \omega_1^M$ places ω_1^M at the corresponding point of the club C . Hence M recovers the appropriate initial part of the odd code $X_{\eta,m,z}$. The transitive collapse of this initial code contains the correct member of $\mathcal{I}_n$ at ω_1^M , the local block and column addresses, and the restrictions of the selected branches. Since $\mathcal{M}$ computes the local Suslin sequence correctly, the collapse verifies that these branches record z . This is precisely the displayed instance of LocalCode_n . $\square$

For later use, let $\dot{Y}_{\eta,m,z}$ denote the $\mathbb{B}_{\eta,m}(z)$ -name for this set.

4.3 The self-indexed coding forcing

For $u \in 2^\omega$, define

$$\Delta(u) = \{2k + 2 : k \in u\} \cup \{2k + 1 : k \notin u\}.$$

Thus $\Delta(u)$ contains exactly one member of each pair

$$\{2k + 1, 2k + 2\}.$$

If $s \in 2^{<\omega}$, put

$$\Delta(s) = \{2k + 2 : k < |s| \text{ and } s(k) = 1\} \cup \{2k + 1 : k < |s| \text{ and } s(k) = 0\}.$$

A condition in $\text{Code}_n(z, \eta)$ is a triple

$$p = (s_p, \pi_p, F_p)$$

with the following properties:

- (1) $s_p \in 2^{<\omega}$;
- (2) π_p is a condition in $\mathbb{B}_\eta^*(z)$ and

$$\text{supp}(\pi_p) \subseteq \Delta(s_p) \times \omega;$$

- (3) F_p is a finite subset of $\omega_1 \times \Delta(s_p)$;
- (4) if $(\xi, m) \in F_p$, then

$$(\pi_p)_m \Vdash_{\mathbb{B}_{\eta, m}(z)} \xi \in \dot{Y}_{\eta, m, z}.$$

Here $(\pi_p)_m$ is the restriction of π_p to column m .

We put $q \leq p$ if:

- (1) $s_p \subseteq s_q$, $\pi_q \leq \pi_p$, and $F_p \subseteq F_q$;
- (2) if $(\xi, m) \in F_p$, then

$$\{j : |s_p| \leq j < |s_q| \text{ and } s_q(j) = 1\} \cap d_{\xi, m} = \emptyset.$$

The fourth clause in the definition of a condition is required for every member of F_q .

Let $K \subseteq \text{Code}_n(z, \eta)$ be generic. Put

$$u = \bigcup \{s_p : p \in K\}.$$

For $m \in \Delta(u)$, let H_m be the filter on $\mathbb{B}_{\eta, m}(z)$ generated by the column conditions $(\pi_p)_m$, $p \in K$.

Lemma 4.3. *The forcing $\text{Code}_n(z, \eta)$ is ccc and has size $\aleph_1$. In its generic extension:*

- (1) if $m \in \Delta(u)$, then H_m is $\mathbb{B}_{\eta, m}(z)$ -generic;
- (2) if $Y_{\eta, m, z}$ is the localized set obtained from H_m , then, for every $\xi < \omega_1$,

$$\xi \in Y_{\eta, m, z} \iff |u \cap d_{\xi, m}| < \omega;$$

- (3) if $m \notin \Delta(u)$, then every tree in column m remains Suslin;
- (4) in an active column $m \in \Delta(u)$, every tree $S_{k,1-z(k)}^{\eta,m}$ remains Suslin.
- (5) every finite product of trees covered by (3) and (4), together with trees outside the block $\vec{S}^\eta$, remains Suslin.

Proof. For the ccc, take an uncountable set of conditions. Pass to an uncountable subset with the same stem s . The branch parts belong to the ccc forcing $\mathbb{B}_\eta^*(z)$, so two of them have a common extension. Since the stems are equal, the union of the two side conditions gives a common extension in $\text{Code}_n(z, \eta)$. The size statement follows from CH.

Fix $m \in \Delta(u)$. Once a stem has decided the bit which selects m , the column remains active. The sets which require the column part to meet a given dense subset of $\mathbb{B}_{\eta,m}(z)$ are dense. Hence H_m is generic.

Suppose that $\xi \in Y_{\eta,m,z}$. By the truth lemma, some column condition in H_m forces this statement. The conditions which then put (ξ, m) into the side condition are dense. After this happens, no later 1 is added on $d_{\xi,m}$. Thus $u \cap d_{\xi,m}$ is finite.

Now suppose that $\xi \notin Y_{\eta,m,z}$. Fix $l < \omega$. Below every condition which has activated column m , the set of extensions whose column part decides the statement

$$\xi \in \dot{Y}_{\eta,m,z}$$

is dense. A condition in the generic filter which decides this statement must force nonmembership. Starting with such a condition, choose

$$j > \max\{l, |s_p|\}$$

in $d_{\xi,m}$ and outside the finitely many sets named by F_p . Extend the stem to length $j+1$, put $s_q(j) = 1$, and put 0 at the other new coordinates. The matrix $\vec{d}$ is pairwise almost disjoint, so this gives a condition. Since l was arbitrary, $u \cap d_{\xi,m}$ is infinite.

It remains to prove the preservation statements. Let T be a finite product of trees not selected by the coding pattern. To show that T remains Suslin, consider an uncountable set of conditions in $\text{Code}_n(z, \eta) \times T$. Pass to a set with one stem. By independence, the corresponding branch parts together with the T -coordinates contain two compatible members. The union of the two side conditions then gives a common extension. Thus $\text{Code}_n(z, \eta) \times T$ is ccc, and T remains Suslin. $\square$

4.4 The coding relation

For a real u and $m \in \Delta(u)$, let

$$Y_m^u = \{\xi < \omega_1 : |u \cap d_{\xi,m}| < \omega\}.$$

If M is a countable transitive model and $\mathcal{M} \in \mathcal{I}_n \cap M$ satisfies $\omega_1^{\mathcal{M}} = \omega_1^M$, define inside M

$$Y_m^{u,\mathcal{M}} = \{\xi < \omega_1^{\mathcal{M}} : |u \cap d_{\xi,m}^{\mathcal{M}}| < \omega\},$$

where $\vec{d}^{\mathcal{M}}$ is the matrix computed from $\mathcal{M}$.

We define $\Theta_n(u, z)$ by

$$\begin{aligned} \Theta_n(u, z) \iff & \forall M \forall \mathcal{M} \left(M \text{ is a countable transitive model of } \text{ZFC}^- + \text{“}\aleph_1 \text{ exists”} \right. \\ & \wedge u, z, \mathcal{M} \in M \\ & \wedge \mathcal{M} \in \mathcal{I}_n \\ & \wedge \omega_1^{\mathcal{M}} = \omega_1^M \\ & \implies \exists \bar{\eta} < \omega_1^M \forall m \in \Delta(u) \ M \models \text{LocalCode}_n(Y_m^{u, \mathcal{M}}, z, \mathcal{M}, \bar{\eta}, m) \Big). \end{aligned}$$

The same local block address $\bar{\eta}$ is required for all active columns.

Lemma 4.4. *The relation $\Theta_n(u, z)$ is Π_{n+2}^1 .*

Proof. The assertion $\mathcal{M} \in \mathcal{I}_n$ is Π_{n+1}^1 . Once M and $\mathcal{M}$ are given, the almost disjoint decoding, the reshaping test, the collapse, and the branch checks in LocalCode_n are first-order checks in the countable coded model M . Thus the matrix of the displayed universal quantification is Σ_{n+1}^1 , and Θ_n is Π_{n+2}^1 . $\square$

Lemma 4.5. *If $\Theta_n(u, z)$ holds, then there is a block address $\eta < \omega_1$ such that, for every $m \in \Delta(u)$ and every $k < \omega$, there is a cofinal branch through*

$$S_{k, z(k)}^{\eta, m}.$$

Proof. Put

$$m_\infty = \bigcup \mathcal{I}_n = \mathcal{J}_{\omega_1}^{M_n}.$$

Fix $m \in \Delta(u)$ and form the global almost disjoint decoding

$$Y_m^u = \{\xi < \omega_1 : |u \cap d_{\xi, m}| < \omega\}.$$

We claim that the canonical localization of Y_m^u succeeds. Suppose otherwise. Then some part of the canonical decoding fails: the reshaping test fails, the odd part does not code a well-founded structure of the required form, or one of the branch clauses fails. Take a countable elementary hull of a sufficiently large H_θ containing u, z, m_∞ , and a witness to the failure, and let M be its transitive collapse. Let $\mathcal{M}$ be the image of m_∞ under the collapse. By the comparison and condensation facts for $\mathcal{I}_n$,

$$\mathcal{M} \in \mathcal{I}_n \quad \text{and} \quad \omega_1^{\mathcal{M}} = \omega_1^M.$$

The almost disjoint decoding computed in M is exactly

$$Y_m^{u, \mathcal{M}} = Y_m^u \cap \omega_1^M.$$

The chosen failure therefore reflects to M . This contradicts $\Theta_n(u, z)$, which says that this uniquely determined set satisfies LocalCode_n in M .

Consequently, Y_m^u decodes a block address η_m and cofinal branches through

$$S_{k, z(k)}^{\eta_m, m} \quad (k < \omega).$$

It remains to see that the address does not depend on m . If two active columns decoded different addresses, take one countable hull containing both canonical decodings and their addresses. In its collapse, the two local decodings still have different addresses. This contradicts the definition of $\Theta_n(u, z)$, which requires one common local address for all members of $\Delta(u)$. Thus there is one $\eta < \omega_1$ which works for every active column. $\square$

Lemma 4.6. *Let z belong to the current intermediate model and let η be fresh. The forcing $\text{Code}_n(z, \eta)$ adds a real u such that $\Theta_n(u, z)$ holds. This remains true after the later forcing used below.*

Proof. Lemma 4.3 gives, for every $m \in \Delta(u)$,

$$Y_m^u = Y_{\eta, m, z}.$$

Let M and $\mathcal{M}$ be as in the definition of $\Theta_n(u, z)$. The correctness of the local almost disjoint matrix gives

$$Y_m^{u, \mathcal{M}} = Y_{\eta, m, z} \cap \omega_1^M.$$

Lemma 4.2 now yields LocalCode_n in every active column, with the same local image of η . Hence $\Theta_n(u, z)$ holds immediately after the coding stage.

The same argument applies to countable transitive models which appear after later forcing. Later coding stages use fresh blocks, and the other iterands preserve the unselected trees in the present block. The real u , its almost disjoint decoding, and the branches already added do not change. Therefore every later suitable model obtains the same localized initial codes, and $\Theta_n(u, z)$ remains true. $\square$

5 The forcing iteration

We now define the final forcing over M_n . Its length is $\omega_1^{M_n}$. We use finite support.

5.1 Bookkeeping

Using the canonical well-order of M_n , fix the least partition

$$\omega_1 = I_{\text{mof}} \cup I_{\text{Am}}$$

into two cofinal sets.

For $X \in \{\text{mof}, \text{Am}\}$, fix a map

$$B_X : I_X \longrightarrow \omega_1 \times \omega_1$$

such that, whenever $B_X(\alpha) = (\beta, \xi)$, we have $\beta < \alpha$, and every pair (β, ξ) occurs cofinally often. Once $\mathbb{P}_\beta$ has been constructed, enumerate its nice names for reals as

$$\langle \dot{r}_\xi^\beta : \xi < \omega_1 \rangle.$$

We allow repetitions if necessary. At a stage $\alpha \in I_X$ with $B_X(\alpha) = (\beta, \xi)$, the bookkeeping presents $\dot{r}_\xi^\beta$, regarded as a $\mathbb{P}_\alpha$ -name. Since $M_n \models \text{CH}$ and $|\mathbb{P}_\beta| \leq \aleph_1$, there are at most $\aleph_1$ such names.

We construct

$$\langle \mathbb{P}_\alpha, \dot{Q}_\alpha : \alpha < \omega_1 \rangle$$

by recursion with finite support. Along the construction we also define $\mathbb{P}_\alpha$ -names

$$\dot{O}_\alpha$$

for the set of non-atomic measures accepted before stage α . Set

$$\mathbb{P}_0 = \mathbf{1} \quad \text{and} \quad \dot{O}_0 = \check{\emptyset}.$$

5.2 The two kinds of stages

Suppose that $\mathbb{P}_\alpha$ and $\dot{\mathcal{O}}_\alpha$ have been defined. Let $G_\alpha \subseteq \mathbb{P}_\alpha$ be generic, and work in $M_n[G_\alpha]$. Write

$$\mathcal{O}_\alpha = \dot{\mathcal{O}}_\alpha^{G_\alpha}.$$

Mof stages. Assume that $\alpha \in I_{\text{mof}}$, and let $\dot{f}_\alpha$ be the name presented by $B_{\text{mof}}(\alpha)$. Put

$$f_\alpha = \dot{f}_\alpha^{G_\alpha}.$$

If $f_\alpha \notin p_c(2^\omega)$, or if μ_{f_α} is not orthogonal to every measure in $\mathcal{O}_\alpha$, let $\mathbb{Q}_\alpha$ be the trivial forcing and set

$$\mathcal{O}_{\alpha+1} = \mathcal{O}_\alpha.$$

Otherwise, call the stage *accepting*. Let η_α be the least block not used at an earlier accepting mof stage, put

$$z_\alpha = \langle \text{mof}, f_\alpha \rangle,$$

and let

$$\mathbb{Q}_\alpha = \text{Code}_n(z_\alpha, \eta_\alpha).$$

Let u_α be the self-indexed almost disjoint real added by this forcing. In the extension by $\mathbb{Q}_\alpha$, put

$$g_\alpha = G_{\text{FT}}(f_\alpha, f_\alpha \oplus u_\alpha).$$

Then

$$\mu_{g_\alpha} \approx \mu_{f_\alpha} \quad \text{and} \quad r_{\text{FT}}(g_\alpha) = f_\alpha \oplus u_\alpha.$$

Set

$$\mathcal{O}_{\alpha+1} = \mathcal{O}_\alpha \cup \{\mu_{g_\alpha}\}.$$

The tag **mof** is used only at accepting mof stages.

Category amoeba stages. If $\alpha \in I_{\text{Am}}$, let $\dot{a}_\alpha$ be the name presented by $B_{\text{Am}}(\alpha)$ and put

$$a_\alpha = \dot{a}_\alpha^{G_\alpha}, \quad N_{a_\alpha} = L[T_{n+1}, a_\alpha].$$

Let $\mathbb{Q}_\alpha$ be the category amoeba forcing computed in N_{a_α} . We use its standard σ -linked presentation, which adds a Borel meager set containing every Borel meager set coded in N_{a_α} . Let $\mathcal{O}_{\alpha+1} = \mathcal{O}_\alpha$.

This description in $M_n[G_\alpha]$ determines a $\mathbb{P}_\alpha$ -name $\dot{\mathbb{Q}}_\alpha$ and a $\mathbb{P}_{\alpha+1}$ -name $\dot{\mathcal{O}}_{\alpha+1}$ in the usual way.

At every successor stage, put

$$\mathbb{P}_{\alpha+1} = \mathbb{P}_\alpha * \dot{\mathbb{Q}}_\alpha.$$

At a limit stage λ , let $\dot{\mathcal{O}}_\lambda$ be the canonical name for

$$\bigcup_{\alpha < \lambda} \dot{\mathcal{O}}_\alpha,$$

and take the finite-support direct limit. Let

$$\mathbb{P} = \mathbb{P}_{\omega_1}.$$

If $G \subseteq \mathbb{P}$ is generic, put

$$V^* = M_n[G] \quad \text{and} \quad \mathcal{O}_{\text{na}} = \bigcup_{\alpha < \omega_1} \mathcal{O}_\alpha.$$

All choices are made with the canonical well-order of M_n , so the recursion defines a forcing $\mathbb{P} \in M_n$.

5.3 Preservation

Lemma 5.1. *For every $\alpha \leq \omega_1$:*

- (1) $\mathbb{P}_\alpha$ is ccc;
- (2) every Suslin tree in a block not used before stage α remains Suslin;
- (3) in a used block, every tree not selected by its coding pattern remains Suslin;
- (4) every finite product of trees covered by (2) and (3) remains Suslin;
- (5) $|\mathbb{P}_\alpha| \leq \aleph_1$.

Proof. The proof is by induction on α . At a mof stage, the interpreted iterand is either trivial or one of the coding forcings from Lemma 4.3. Hence it is ccc and preserves every finite product outside the active coding coordinates. The category amoeba forcing used here is σ -linked, so it preserves every Suslin tree, including each finite product occurring in (4). Finite-support iteration preserves the stated properties. The size bound follows from CH in M_n and the fact that every iterand has size at most $\aleph_1$. $\square$

Lemma 5.2. *Every real in V^* belongs to some intermediate extension $M_n[G_\beta]$. Moreover, if $\dot{r}$ is a nice $\mathbb{P}_\beta$ -name for a real, then cofinally many later stages in I_{mof} present $\dot{r}$, and cofinally many later stages in I_{Am} present $\dot{r}$.*

Proof. Since $\mathbb{P}$ is ccc, a nice name for a real has countable support. This support is bounded in ω_1 , so the name is already a $\mathbb{P}_\beta$ -name for some $\beta < \omega_1$. It occurs in the canonical enumeration of the nice $\mathbb{P}_\beta$ -names. The conclusion follows from the defining property of B_{mof} and B_{Am} . $\square$

Corollary 5.3. *The forcing $\mathbb{P}$ preserves all cardinals and*

$$V^* \models \text{CH}.$$

Moreover, $\mathbb{P}$ has M_n -cardinality $\omega_1 < \delta_0$. Hence V^ is a small generic extension of M_n , and the projective correctness and comparison facts from Theorem 3.2 hold in V^* .*

Proof. The forcing is ccc and has size $\aleph_1$. Since $M_n \models \text{CH}$, there are only $\aleph_1$ nice names for reals. Thus $V^* \models 2^{\aleph_0} = \aleph_1$. The final statement follows from $\omega_1 < \delta_0$ and Theorem 3.2. $\square$

5.4 Exactness of the reserved code

Lemma 5.4. *In V^* , suppose that*

$$\Theta_n(u, \langle \mathbf{mof}, f \rangle)$$

holds. Then there is a unique accepting mof stage α such that

$$z_\alpha = \langle \mathbf{mof}, f \rangle \quad \text{and} \quad u = u_\alpha.$$

In particular, $f_\alpha = f$, and stage α used block η_α to code this real.

Proof. By Lemma 4.5, the relation $\Theta_n(u, \langle \mathbf{mof}, f \rangle)$ yields a block η such that every column in $\Delta(u)$ contains the branch pattern for $\langle \mathbf{mof}, f \rangle$. An unused block has no branch through any of its trees. Hence $\eta = \eta_\alpha$ for an accepting mof stage α . Since every accepting stage uses a fresh block, this stage is determined by η .

Let z_α be the real coded at that stage. The active columns of the block are exactly the columns in $\Delta(u_\alpha)$. Every column outside $\Delta(u_\alpha)$ remains Suslin by Lemma 5.1. Since the decoded pattern gives branches in every column in $\Delta(u)$, we have

$$\Delta(u) \subseteq \Delta(u_\alpha).$$

Each of these two sets contains exactly one member of every pair $\{2k+1, 2k+2\}$. Therefore

$$\Delta(u) = \Delta(u_\alpha),$$

and hence $u = u_\alpha$.

It remains to identify the real written into the block. If

$$z_\alpha \neq \langle \mathbf{mof}, f \rangle,$$

choose $k < \omega$ at which the two reals differ. In every active column the stage added a branch through the tree selected by $z_\alpha(k)$, while Lemma 4.5 gives a branch through the other member of the pair, selected by $\langle \mathbf{mof}, f \rangle(k)$. The latter tree remains Suslin by Lemma 5.1, a contradiction. Thus

$$z_\alpha = \langle \mathbf{mof}, f \rangle.$$

Since $z_\alpha = \langle \mathbf{mof}, f_\alpha \rangle$, we have $f_\alpha = f$. Once the stage has added a measure equivalent to μ_f , no later stage which presents f can be accepting. This proves the asserted uniqueness. $\square$

6 The Baire property

We now prove the regularity statement needed for the negative mof argument. Of the forcing notions occurring in the iteration, only category amoeba forcing is used.

For a real $a \in V^*$, let

$$N_a = L[T_{n+1}, a].$$

The model N_a is the base for the auxiliary Cohen-forcing argument and the category-covering argument.

Lemma 6.1 (Category covering). *For every real $a \in V^*$, there is a Borel meager set $Z_a \in V^*$ such that every Borel meager set coded in N_a is contained in Z_a . Consequently, the set of reals which are not Cohen-generic over N_a is meager.*

Proof. By Lemma 5.2, the real a belongs to some intermediate extension and its name is presented at a later category amoeba stage. Let Z_a be the Borel meager set added at that stage. By the definition of category amoeba forcing, every Borel meager set coded in N_a is contained in Z_a . This inclusion is preserved by later forcing.

A real is not Cohen-generic over N_a if and only if it belongs to a Borel meager set coded in N_a . Hence every such real belongs to Z_a . $\square$

Lemma 6.2 (Projective correctness). *For all reals $a, z \in V^*$,*

$$N_a[z] = L[T_{n+1}, a, z] = L[T_{n+1}, a \oplus z] \prec_{\Sigma_{n+2}^1} V^*.$$

Consequently, if $\varphi(v, a)$ is Σ_{n+2}^1 , then

$$N_a[z] \models \varphi(z, a) \iff V^* \models \varphi(z, a).$$

Proof. By Corollary 5.3, V^* is an M_n -generic extension by a forcing of M_n -cardinality below δ_0 . Apply Theorem 3.2(3) to the real $a \oplus z$. $\square$

Theorem 6.3. *In V^* , every boldface Σ_{n+2}^1 set of reals has the Baire property.*

Proof. Let $A \subseteq \omega^\omega$ be $\Sigma_{n+2}^1(a)$, say

$$x \in A \iff \varphi(x, a),$$

where φ is Σ_{n+2}^1 , and put

$$N_a = L[T_{n+1}, a].$$

Work in N_a with Cohen forcing, and let $\dot{c}$ be its canonical name. The Boolean value of $\varphi(\dot{c}, a)$ is represented, modulo meager sets, by a Borel set C coded in N_a .

Let z be Cohen-generic over N_a . The forcing theorem gives

$$z \in C \iff N_a[z] \models \varphi(z, a).$$

By Lemma 6.2, therefore,

$$z \in C \iff z \in A$$

for every Cohen real z over N_a .

It follows that $A \triangle C$ is contained in the set of reals which are not Cohen-generic over N_a . This set is meager by Lemma 6.1. Hence A has the Baire property. $\square$

7 Σ_{n+2}^1 -uniformization in small extensions of M_n

The uniformization conclusion is not supplied by the forcing iteration. It follows from Steel's analysis of the mice $M_n(s)$ in small generic extensions of M_n . We state the form of that analysis which is used here; a proof in the present notation is given in the two companion papers [Hof26, HM26, for the Π -side, see Proposition 5.12 and Theorem 5.13]. Throughout this section, a *small generic extension* means an extension by set forcing in M_n of M_n -cardinality below δ_0 , in the class considered here, so that the canonical M_n -strategy and the relevant comparison arguments are preserved.

Let

$$N = M_n[H],$$

be such a small generic extension. Fix $s \in \mathbb{R}^N$, and let $M_n(s)$ be the canonical proper class s -mouse with n Woodin cardinals. In N , let $\mathcal{I}_n(s)$ be the class of codes for countable s -premise $\mathcal{P}$ such that:

- (1) $\mathcal{P}$ is sound and n -small over s ;
- (2) $\rho_\omega(\mathcal{P}) = \omega$;
- (3) $\mathcal{P}$ satisfies Steel's Π_n -iterability condition.

The relation

$$\mathcal{P} \in \mathcal{I}_n(s)$$

is $\Pi_{n+1}^1(s)$ in codes. Comparison linearly orders the members of $\mathcal{I}_n(s)$ by initial segment and identifies them with a cofinal system of countable initial segments of $M_n(s)$.

Proposition 7.1. *Let $p \in \mathbb{R} \cap M_n(s)$, and let $\theta(s, p, u, v)$ be Π_{n+1}^1 . Then:*

- (1) *if*

$$N \models \exists u \exists v \theta(s, p, u, v),$$

then there are $u, v \in \mathbb{R} \cap M_n(s)$ such that

$$M_n(s) \models \theta(s, p, u, v);$$

- (2) *for $u, v \in \mathbb{R} \cap M_n(s)$,*

$$M_n(s) \models \theta(s, p, u, v) \iff N \models \theta(s, p, u, v);$$

- (3) *the canonical construction order $<_{n,s}$ on $\mathbb{R} \cap M_n(s)$ is a good $\Sigma_{n+2}^1(s)$ well-order. In particular, the relation*

$$\text{Least}_\theta^n(s, p, u)$$

saying that u is the $<_{n,s}$ -least real in $M_n(s)$ for which

$$M_n(s) \models \exists v \theta(s, p, u, v)$$

is uniformly $\Sigma_{n+2}^1(s, p)$.

Proof. The complexity calculation for $\mathcal{I}_n(s)$ is the relativized version of Theorem 3.2(1). Soundness, n -smallness, and the projectum condition are arithmetical in a premouse code; the only projective part is the Π_n -iterability requirement, which is $\Pi_{n+1}^1(s)$. The comparison theorem therefore identifies the correctly iterable codes without increasing the projective level.

For (1), suppose that the displayed $\Sigma_{n+2}^1(s, p)$ statement holds in N . Steel's comparison-and-capture argument places witnesses into a countable member of $\mathcal{I}_n(s)$. Otherwise, comparison with the canonical s -mouse construction would produce an iterable n -small s -premouse which is not an initial segment of $M_n(s)$, contrary to comparison. This gives $u, v \in M_n(s)$ with the required internal statement. The same comparison, applied to a putative disagreement about a Π_{n+1}^1 statement with parameters in $M_n(s)$, gives (2). Thus $M_n(s)$ is Σ_{n+2}^1 -correct in N for real parameters which belong to it.

For (3), assign to a real $u \in M_n(s)$ the least member of the comparison order which contains u , and then use the construction order of that premouse. Coherence under comparison makes the

resulting relation independent of the chosen code and gives a well-order $<_{n,s}$. Since membership in $\mathcal{I}_n(s)$ is $\Pi_{n+1}^1(s)$, a code for the relevant initial segment is an existential real witness over a Π_{n+1}^1 predicate. Hence the initial segments of $<_{n,s}$ are uniformly $\Sigma_{n+2}^1(s)$. The same calculation, with the Π_{n+1}^1 matrix θ , gives the displayed least-witness relation. $\square$

Theorem 7.2. *Every small generic extension of M_n in the preceding sense satisfies boldface Σ_{n+2}^1 -uniformization. In particular, V^* satisfies boldface Σ_{n+2}^1 -uniformization.*

Proof. Work in such an extension N . Let $A \subseteq \mathbb{R}^2$ be $\Sigma_{n+2}^1(a)$. Fix a Π_{n+1}^1 formula ψ such that

$$A(x, y) \iff \exists z \psi(a, x, y, z).$$

For each $x \in \mathbb{R}$, put

$$s = a \oplus x$$

and define, inside $M_n(s)$,

$$C_{a,x} = \{y \in \mathbb{R} \cap M_n(s) : M_n(s) \models \exists z \psi(a, x, y, z)\}.$$

Now set

$$U(x, y) \iff y \in C_{a,x} \text{ and } y \text{ is } <_{n,a \oplus x} \text{-least in } C_{a,x}.$$

After decoding a and x from $s = a \oplus x$, this is an instance of the least-witness relation in Proposition 7.1(3). Therefore

$$U \in \Sigma_{n+2}^1(a).$$

If $U(x, y)$ holds, choose $z \in M_n(a \oplus x)$ such that

$$M_n(a \oplus x) \models \psi(a, x, y, z).$$

By Proposition 7.1(2), $N \models \psi(a, x, y, z)$, so $A(x, y)$ holds. Hence $U \subseteq A$. The relation U is single-valued because $<_{n,a \oplus x}$ is a well-order.

Finally, suppose that the x -section of A is nonempty. Then

$$N \models \exists y \exists z \psi(a, x, y, z).$$

Proposition 7.1(1), applied with $s = a \oplus x$, gives witnesses $y, z \in M_n(a \oplus x)$. Thus $C_{a,x}$ is nonempty and has a $<_{n,a \oplus x}$ -least member. Consequently the x -section of U is nonempty. Therefore U uniformizes A .

By Corollary 5.3, the final model V^* is such a small generic extension of M_n . $\square$

8 The Π_{n+2}^1 maximal orthogonal family

We now verify maximality and definability of the family built during the iteration.

Lemma 8.1. *In V^* , the family $\mathcal{O}_{na}$ is a maximal orthogonal family of non-atomic measures.*

Proof. Suppose that μ_{g_α} is added at stage α . At that stage, μ_{f_α} is orthogonal to every earlier member of the family. Since

$$\mu_{g_\alpha} \approx \mu_{f_\alpha},$$

the new measure is orthogonal to every earlier member. Later accepting stages test orthogonality against the enlarged family. Thus the family is pairwise orthogonal.

For maximality, suppose that $h \in p_c(2^\omega)$ and μ_h is orthogonal to every member of $\mathcal{O}_{\text{na}}$. The real h appears in some intermediate extension. At a later mof stage, the bookkeeping presents a name whose value is h . Since μ_h is orthogonal to the final family, it is orthogonal to the family constructed up to that stage. The stage therefore accepts h and adds a measure equivalent to μ_h , contradicting the assumption that μ_h is orthogonal to every member of the final family. $\square$

For $g \in \text{dom}(r_{\text{FT}})$, write

$$r_{\text{FT}}(g) = f_g \oplus u_g$$

using the fixed recursive projections.

Theorem 8.2. *The set of codes for the measures in $\mathcal{O}_{\text{na}}$ is lightface Π_{n+2}^1 .*

Proof. Define $g \in \widehat{\mathcal{O}}_{\text{na}}$ if and only if:

- (1) $g \in p_c(2^\omega) \cap \text{dom}(r_{\text{FT}})$;
- (2) $f_g \in p_c(2^\omega)$;
- (3) $g = G_{\text{FT}}(f_g, f_g \oplus u_g)$;
- (4) $\Theta_n(u_g, \langle \text{mof}, f_g \rangle)$.

The first three clauses are arithmetical. The last clause is Π_{n+2}^1 by Lemma 4.4. Hence $\widehat{\mathcal{O}}_{\text{na}}$ is Π_{n+2}^1 .

If $g = g_\alpha$ was added at an accepting mof stage, then

$$r_{\text{FT}}(g) = f_\alpha \oplus u_\alpha, \quad g = G_{\text{FT}}(f_\alpha, f_\alpha \oplus u_\alpha),$$

and $\Theta_n(u_\alpha, \langle \text{mof}, f_\alpha \rangle)$ holds. Thus $g \in \widehat{\mathcal{O}}_{\text{na}}$.

Conversely, suppose that $g \in \widehat{\mathcal{O}}_{\text{na}}$. By Lemma 5.4, there is a unique accepting mof stage which coded $\langle \text{mof}, f_g \rangle$, and its coding real is u_g . The measure code produced at that stage is

$$G_{\text{FT}}(f_g, f_g \oplus u_g) = g.$$

Thus g was added to the family. Therefore

$$\widehat{\mathcal{O}}_{\text{na}} = \{g : \mu_g \in \mathcal{O}_{\text{na}}\}.$$

$\square$

Let

$$\mathcal{O} = \mathcal{O}_{\text{na}} \cup \{\delta_x : x \in 2^\omega\}.$$

The set of point measures is Borel. Every point measure is orthogonal to every non-atomic measure, and distinct point measures are orthogonal. A measure which is orthogonal to every point measure has no atoms. Lemma 8.1 therefore implies that $\mathcal{O}$ is maximal in the full space $P(2^\omega)$.

Corollary 8.3. *In V^* , the family $\mathcal{O}$ is a lightface Π_{n+2}^1 maximal orthogonal family in $P(2^\omega)$.*

9 The absence of Σ_{n+2}^1 maximal orthogonal families

We now prove the negative result used in the main theorem.

Proposition 9.1. *Assume that every boldface Σ_{n+2}^1 set of reals has the Baire property and that every total boldface Σ_{n+2}^1 relation on the reals has a boldface Σ_{n+2}^1 uniformization. Then there is no boldface Σ_{n+2}^1 maximal orthogonal family of Borel probability measures on 2^ω .*

Proof. Suppose that $A \subseteq P(2^\omega)$ is a $\Sigma_{n+2}^1(a)$ maximal orthogonal family. Let $x \mapsto \mu^x$ be the Kechris–Sofronidis map for E_I , so that

$$xE_I y \Rightarrow \mu^x \approx \mu^y, \quad \neg xE_I y \Rightarrow \mu^x \perp \mu^y.$$

For $x \in 2^\omega$, put

$$A_x = \{\nu \in A : \nu \not\perp \mu^x\}.$$

Maximality gives $A_x \neq \emptyset$. Moreover, A_x is countable. Indeed, for each $\nu \in A_x$ choose a nonzero probability measure η_ν such that

$$\eta_\nu \ll \nu \quad \text{and} \quad \eta_\nu \ll \mu^x.$$

The measures η_ν , $\nu \in A_x$, are pairwise orthogonal and all lie below μ^x ; hence the ccc of the measure algebra below μ^x implies that A_x is countable.

Define $Q(x, (\nu_i)_{i < \omega})$ by

$$\begin{aligned} Q(x, (\nu_i)_{i < \omega}) \iff & (\forall i < \omega)(\nu_i \in A \wedge \nu_i \not\perp \mu^x) \\ & \wedge (\forall \rho \in P(2^\omega))(\rho \not\perp \mu^x \rightarrow (\exists i < \omega) \nu_i \not\perp \rho). \end{aligned} \quad (\dagger)$$

Every section Q_x is nonempty. To see this, enumerate A_x as $(\nu_i)_{i < \omega}$. If $\rho \not\perp \mu^x$, choose a nonzero probability measure η such that $\eta \ll \rho$ and $\eta \ll \mu^x$. By maximality, some $\nu \in A$ satisfies $\nu \not\perp \eta$. Hence $\nu \not\perp \rho$ and $\nu \not\perp \mu^x$, so $\nu \in A_x$ and $(\dagger)$ holds. Conversely, if $Q(x, (\nu_i))$ and $\nu \in A_x$, apply the second clause with $\rho = \nu$. Since A is orthogonal, $\nu_i \not\perp \nu$ implies $\nu_i = \nu$. Thus

$$Q(x, (\nu_i)) \implies \{\nu_i : i < \omega\} = A_x.$$

The relations $\perp$ and $\ll$ are arithmetical on measure codes. Hence the first clause of $(\dagger)$ is $\Sigma_{n+2}^1(a)$, while the second is Π_1^1 . Therefore Q is $\Sigma_{n+2}^1(a)$. By uniformization, there is a total function

$$F : 2^\omega \longrightarrow P(2^\omega)^\omega$$

with $\Sigma_{n+2}^1(a)$ graph and $Q(x, F(x))$ for every x . For every open $U \subseteq P(2^\omega)^\omega$,

$$F^{-1}(U) = \{x : (\exists \vec{\nu}) [\text{Graph}(F)(x, \vec{\nu}) \wedge \vec{\nu} \in U]\}$$

is $\Sigma_{n+2}^1(a)$, hence has the Baire property. Therefore F is Baire measurable.

If $xE_I y$, then $\mu^x \approx \mu^y$, and consequently

$$\{F(x)(i) : i < \omega\} = A_x = A_y = \{F(y)(i) : i < \omega\}.$$

Define

$$\tilde{F}(x)(\langle i, j \rangle) = F(x)(i).$$

Then $\tilde{F}$ is Baire measurable and

$$xE_I y \implies \tilde{F}(x) E_{S_\infty} \tilde{F}(y),$$

where E_{S_∞} is the coordinate-permutation orbit relation. By Hjorth turbulence, there are a comeager set $D \subseteq 2^\omega$ and a fixed nonempty countable set $C \subseteq P(2^\omega)$ such that

$$A_x = C \quad (x \in D).$$

Since every E_I -class is meager, choose pairwise E_I -inequivalent reals $x_\alpha \in D$, $\alpha < \omega_1$. Fix $\nu \in C$. Then $\nu \not\ll \mu^{x_\alpha}$ for every $\alpha < \omega_1$. Choose nonzero probability measures η_α such that

$$\eta_\alpha \ll \nu \quad \text{and} \quad \eta_\alpha \ll \mu^{x_\alpha}.$$

The measures $(\eta_\alpha)_{\alpha < \omega_1}$ are pairwise orthogonal and all lie below ν , contradicting the ccc of the measure algebra below ν . Hence no such A exists. $\square$

10 Proof of the main theorem

Proof of Theorem 1.1. Let $V^* = M_n[G]$ be the extension by the iteration from Section 5. Corollary 5.3 gives cardinal preservation and CH. Theorem 6.3 gives the Baire property for all boldface Σ_{n+2}^1 sets. Theorem 7.2 gives boldface Σ_{n+2}^1 -uniformization. Corollary 8.3 gives a lightface Π_{n+2}^1 mof. Proposition 9.1, applied to the preceding two conclusions, rules out every boldface Σ_{n+2}^1 mof. $\square$

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